1 **(a)** Boys Total = 40 and Girls Football = 8 **[B1]**

Girls Basketball = 8 and Total Tennis = 27 **[B1]**

Boys Tennis = 13 **[B1]**

(b) $\frac{13}{70}$ **[B1]**

[Total Marks: 4]

2 **(a)** Tennis Total = 15, Girls Total = 22 **[B1]**

Boys Football = 12, Boys Swimming = 9 **[B1]**

Girls Football = 8 **[B1]**

(b) $\frac{12}{50}$ or $\frac{6}{25}$ **[B1]**

[Total Marks: 4]

3 **(a)** 5 in Boys/Swimming cell **[B1]**

5 in Girls/Football, 9 in Girls/Swimming, 22 in Girls/Total **[B1]**

16 in Total/Tennis, 14 in Total/Swimming **[B1]**

(b) $\frac{9}{50}$ **[B1]**

[Total Marks: 4]

4 **(a)** Boys Basketball = 8 **[B1]**

Girls Football = 6 AND Girls Total = 22 **[B1]**

Total Basketball = 18 AND Total Tennis = 11 **[B1]**

(b) Denominator of 11 seen or fraction $\frac{5}{n}$ where $n > 5$ **[M1]**

$\frac{5}{11}$ **[A1]**

[Total Marks: 5]

5 **(a)** 6 or more correct missing entries **[B3]**

4 or 5 correct missing entries **[B2]**

2 or 3 correct missing entries **[B1]**

(Correct Boys row: 24, 10, 11, 45)

(Correct Girls row: 14, 12, 9, 35)

(Correct Total row: 38, 22, 20, 80)

(b) $\frac{10}{80}$ or $\frac{1}{8}$ **[B1]**

(c) Denominator of 35 seen **[M1]**

$\frac{9}{35}$ **[A1]**

[Total Marks: 6]

6 (a) (i) Qualitative **[B1]**

(ii) Continuous **[B1]**

(iii) Discrete **[B1]**

(b) Row $60 \leq r < 70$: correct tally of 8 and frequency 8 **[B1]**

Row $70 \leq r < 80$: correct tally of 8 and frequency 8 **[B1]**

Row $80 \leq r < 90$: correct tally of 4 and frequency 4 **[B1]**

(c) $\frac{4+8}{24} \times 100$ **[M1]**

50 **[A1]**

[Total Marks: 8]

7 (a) $x + (2x + 10) + (x + 5) + (3x - 11)$ **[M1]**

$7x + 4$ **[A1]**

(b) $7x + 4 = 130$ **[M1]**

$x = 18$ **[A1]**

(c) Correct Male row: 18, 46, 64 **[B1]**

Correct Female row: 23, 43, 66 **[B1]**

Correct Total row: 41, 89, 130 **[B1]**

(d) Denominator of 41 seen **[M1]**

$\frac{23}{41}$ **[A1]**

[Total Marks: 9]

8 (a) (i) $(2x + y) + (x + 6) = 28$ **[M1]**

$22 - 3x$ **[A1]**

(ii) $(x^2 + 2x) + (2x + y) + (4x - y) + (2x + 3y) + (x + 6) + (3x + 3y) = 150$ simplifying to $x^2 + 14x + 6y + 6 = 150$ **[M1]**

$x^2 + 14x + 6(22 - 3x) - 144 = 0$ **[M1]**

$x^2 + 14x + 132 - 18x - 144 = 0$ simplifying to $x^2 - 4x - 12 = 0$ **[A1]**

(b) $(x - 6)(x + 2) = 0$ **[M1]**

$x = 6$ (rejecting $x = -2$ since $x > 0$) **[A1]**

$y = 22 - 3(6) = 4$ **[A1]**

(c) **(i)** Row *Yes*: $48, 16, 20$, Total $= 84$ **[B1]**

Row *No*: $24, 12, 30$, Total $= 66$; Column Totals: $72, 28, 50$ **[B1]**

(ii) $\frac{84}{150} \times 100$ **[M1]**

56 **[A1]**

[Total Marks: 12]

9 **(a)** Total Women $= 55$, Total Tea $= 70$, Overall Total $= 120$ placed correctly **[B1]**

Men Coffee $= 25$ and Total Coffee $= 50$ or Men Total $= 65$ placed correctly **[B1]**

Men Tea $= 40$, Women Tea $= 30$, Women Coffee $= 25$ **[B1]**

(b) $\frac{30}{120} \times 100$ **[M1]**

25 **[A1]**

[Total Marks: 5]

10 **(a)** Total column correct: 75 (Men), 45 (Women), 120 (Total) **[B1]**

Bus column correct: 15 (Men), 15 (Women), 30 (Total) **[B1]**

Train column correct: 24 (Men), 14 (Women), 38 (Total) **[B1]**

Car column correct: 36 (Men), 16 (Women), 52 (Total) **[B1]**

(b) Denominator of 52 seen or fraction $\frac{36}{n}$ **[M1]**

$\frac{36}{52}$ or $\frac{9}{13}$ **[A1]**

[Total Marks: 6]

11 **(a)** Total Juniors $= 240$, Total Seniors $= 160$ **[B1]**

Total Bus $= 180$, Total Car $= 110$, Total Walk $= 110$ **[B1]**

Juniors: Bus $= 100$, Car $= 80$, Walk $= 60$ AND Seniors: Bus $= 80$, Car $= 30$, Walk $= 50$ **[B1]**

(b) $\frac{60}{400}$ or $\frac{3}{20}$ **[B1]**

(c) $\frac{80}{160} \times \frac{79}{159}$ **[M1]**

$\frac{79}{318}$ **[A1]**

[Total Marks: 6]

12 **(a)** $x + 2x - 5 + x + 6 + x - 2 + x + 5 + 2x + 1 = 85$ or $8x + 5 = 85$ **[M1]**

$8x = 80$ **[M1]**

10 **[A1]**

(b) Man who prefers Water $= 10 + 6 = 16$ **[M1]**

$\frac{16}{85}$ **[A1]**

(c) Total who prefer Coffee $= 15 + 15 = 30$ **[M1]**

$\frac{15}{30}$ or $\frac{1}{2}$ **[A1]**

[Total Marks: 7]

13 (a) Row totals: Year 9 = 96, Year 10 = 114, Year 11 = 90 **[B1]**

Column total Bus = 100 **[B1]**

Year 9 row completed correctly: Walk = 35, Bus = 36, Car = 25 **[B1]**

Year 11 row completed correctly: Walk = 30, Bus = 42 **[B1]**

Year 10 row completed correctly: Walk = 50, Bus = 22, Car = 42 **[B1]**

(b) $\frac{42}{114} \times 100$ $(= 36.842\ldots)$ **[M1]**

36.8 **[A1]**

(c) $\frac{42}{100}$ **[M1]**

$\frac{21}{50}$ **[A1]**

[Total Marks: 9]

14 (a) $x + 2x + (x + 5) + (3x - 2) = 80$ **[M1]**

$7x + 3 = 80$ **[M1]**

$x = 11$ **[A1]**

(b) Boys Bus = 11, Boys Walk = 22, Boys Total = 33 **[B1]**

Girls Bus = 16, Girls Walk = 31, Girls Total = 47 and Column totals (Bus = 27, Walk = 53) correct **[B1]**

(c) Denominator of 53 seen **[M1]**

$\frac{31}{53}$ **[A1]**

[Total Marks: 7]

15 (a) Boys total = 120 and Girls total = 180 **[B1]**

Boys/Art = 24 **[B1]**

Boys/Drama = 48 and Boys/Music = 48 **[B1]**

Girls/Drama = 60 **[B1]**

Girls/Art = 45 and Girls/Music = 75 AND all totals correct $(69, 123, 108, 300)$ **[B1]**

(b) Girls who do not choose Drama $= 45 + 75 = 120$ **[M1]**

$\frac{120}{300}$ or $\frac{2}{5}$ **[A1]**

[Total Marks: 7]

16 (a) Correct Managers row: 48, 12, 60 **[B1]**

Correct Office Workers row: 90, 60, 150 **[B1]**

Correct Technicians row: 78, 12, 90 **[B1]**

Correct Total row: 216, 84, 300 **[B1]**

(b) Denominator of 84 seen **[M1]**

$\frac{60}{84}$ or $\frac{5}{7}$ **[A1]**

(c) $\frac{12}{60} \times \frac{48}{59}$ **[M1]**

$\frac{12}{60} \times \frac{48}{59} + \frac{48}{60} \times \frac{12}{59}$ or $2 \times \left(\frac{12}{60} \times \frac{48}{59}\right)$ **[M1]**

$\frac{96}{295}$ or 0.325 (0.3254...) **[A1]**

[Total Marks: 9]

17 (a) $\frac{3}{4}x$ or $0.75x$ **[B1]**

(b) Forms correct equation: $\frac{3}{4}x + \left(\frac{1}{4}x + 30\right) = 110$ **[M1]**

Simplifies to $x + 30 = 110$ leading to $x = 80$ with no errors **[A1]**

(c) Male row correct: 20 (Golf), 60 (Tennis), 80 (Total) **[B1]**

Female row correct: 70 (Golf), 50 (Tennis), 120 (Total) **[B1]**

Total row correct: 90 (Golf), 110 (Tennis), 200 (Total) **[B1]**

(d) Identifies correct fraction for first female: $\frac{70}{90}$ **[M1]**

Multiplies by correct fraction for second female: $\frac{70}{90} \times \frac{69}{89}$ **[M1]**

$\frac{4830}{8010}$ or $\frac{161}{267}$ **[A1]**

[Total Marks: 9]

18 (a) Total number of left-handed students $= 3x$ **[B1]**

$\frac{3x}{120} \times \frac{3x-1}{119} = \frac{3}{34}$ **[M1]**

$\frac{9x^2 - 3x}{14280} = \frac{3}{34}$ or $3x(3x-1) = 1260$ **[M1]**

$9x^2 - 3x = 1260 \implies 3x^2 - x - 420 = 0$ (with no errors seen) **[A1]**

(b) $(3x - 36)(x + 11.66...) = 0$ or correct substitution into quadratic formula **[M1]**

$x = 12$ **[A1]**

Total students equation: $x + y + 2x + y - 6 = 120$ **[M1]**

$3(12) + 2y - 6 = 120 \implies 2y = 90 \implies y = 45$ **[A1]**

(c) Total right-handed $= 2y - 6 = 2(45) - 6 = 84$ OR Right-handed girls $= 45 - 6 = 39$ **[M1]**

$\frac{39}{84}$ or $\frac{13}{28}$ **[A1]**

[Total Marks: 10]